

Demonstrating the Ratio of Time Dilation Between Two Systems Emerges Solely From Their Relative Respective Mass to Radius Ratios

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ABSTRACT: General relativity tells us that time is relativistic; it only has meaning when compared to another system. However, in the standard interpretation of general relativity, it feels very much like one can simply calculate the relativistic time dilation for a single system. That is in fact a misconception; if you look carefully at the Schwarzschild threshold condition for gravitational collapse, it is actually comparing two systems: one, the stated system being evaluated, and two, that same system at its Schwarzschild radius according to its mass. A gravitational time dilation figure is returned; that value is mathematically identical to the ratio of the gravitational time dilation of one system to the gravitational time dilation of the other. We demonstrate and validate this principle for the above example in addition to four other distinct cases that span the classical to relativistic regimes: comparing two systems of constant radius, comparing two systems of constant inertia, comparing two systems of constant density, and comparing two systems of distinct mass and radius.

Keywords: time dilation, general relativity, special relativity, gravitational time dilation, Lorentz time dilation, DeGerlia compactness, inertial density, moment of inertia, Schwarzschild radius

1 Introduction

The Schwarzschild gravitational time-dilation calculation $\sqrt{1-r_s/R}$ contains two radii: R , the radius of a system, and $r_s = 2GM/c^2$, the Schwarzschild radius of the same mass. Same mass at two different radii: the formula compares two systems. The notation places r_s inside one symbol, which gives the formula a single-system appearance, but the calculation is two-system. The single-system reading gives the right number — the formula carries both systems — and is incoherent within relativity, where a frame of reference requires a comparator.

This paper writes both systems openly. For each uniform spherical system, the DeGerlia compactness $D = M/R$ (kg/m) is the mass-to-radius ratio¹. The DeGerlia compactness ratio $k_d = D_1/D_2 = M_1R_2/(M_2R_1)$ governs the time dilation between two such systems. The static-mass case ($M_1 = M_2$, different radii) is the standard gravitational time-dilation calculation^{2,3}. The static-radius case ($R_1 = R_2$, different masses) is the standard Lorentz time-dilation calculation⁴. Six worked cases in Section 5 and the appendix tables (Appendix Appendix A) tabulate the DeGerlia compactness ratio k_d and the gravitational time-dilation calcu-

lation for each pair across the classical and relativistic regimes. Agreement is not asserted on trust: every input and output is tabulated in the appendix, and each row is reproducible directly from $r_s = 2GM/c^2$ and the standard Schwarzschild form.

2 General Relativity and Compactness

2.1 Compactness

Compactness is the dimensionless ratio of a system's mass to its radius, scaled by G/c^2 . The most common forms are $C = GM/(Rc^2)$ and $r_s/R = 2GM/(Rc^2) = 2C$, where $r_s = 2GM/c^2$ is the Schwarzschild radius. Both measure how close a system sits to its Schwarzschild threshold: $r_s/R = 1$ marks the gravitational-collapse condition.

2.2 Inertial Density P

Inertial density P^1 is the moment of inertia of a system divided by its volume:

$$P = I/V \quad (1)$$

Since linear inertia is mass, this reduces directly to mass density $\rho = M/V$ in the linear case; inertial density is the rotational generalization of mass density, a quantity already familiar from linear mechanics. Or more accurately, mass density is simply the linear

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interpretation of inertial density.

Substituting $I = \frac{2}{5}MR^2$ and $V = \frac{4}{3}\pi R^3$, with R the radius, this reduces to:

$$P = \frac{M}{R} \times \frac{3}{10\pi} \quad (2)$$

Just as ordinary density ($\rho = M/V$) is mass distributed over volume, inertial density is rotational inertia distributed over volume. As stated in the prior work: *mass and density are to linear motion as moment of inertia and inertial density are to rotational motion.*

2.3 DeGerlia Compactness D

$D = M/R$ is a simplified analog to inertial density, related to P by the geometric factor $3/10\pi$, and represents the mass over radius ratio for a system. Given that this quantity is almost always used in a ratio that would cancel out all geometric vectors and constants, for ease of use, this metric D for inertial density is typically used over the geometrically accurate inertial density, P .

2.4 DeGerlia Threshold D_{crit}

The Schwarzschild condition can be expressed as a static universal constant threshold of DeGerlia compactness D . There are several advantages to this approach, not the least of which is that you don't have to calculate the threshold for every mass. And, it's much easier to calculate D than P . Inertial density is an intuitive metric that's easier to work with.

Setting $R = r_s$ (the Schwarzschild condition)³ and solving yields the **DeGerlia Threshold**: the universal constant of DeGerlia compactness at which the escape velocity equals the speed of light:

$$D_{crit} = \frac{c^2}{2G} = (6.73295 \pm 0.00015) \times 10^{26} \text{ kg/m} \quad (3)$$

Note this value is exact; its precision is limited by $G = (6.67430 \pm 0.00015) \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$.

3 The Space-Time Equivalence

The following identity relates the time-dilation ratio of two uniform spherical systems to their mass and radius.*

$$k_d = \frac{I_1}{I_2} \cdot \left(\frac{R_2}{R_1} \right)^3 = \frac{D_1}{D_2} = \frac{M_1 R_2}{M_2 R_1} \quad (4)$$

where I_1, I_2 are the moments of inertia, R_1, R_2 are the radii, M_1, M_2 are the masses, and $D_i = M_i/R_i$ is the DeGerlia Compactness of system i . The three forms in Equation 4 are algebraically identical; $(I_1/I_2)(R_2/R_1)^3$ exposes the inertia-radius structure, D_1/D_2 exposes the compactness-ratio structure, and $M_1 R_2/(M_2 R_1)$ is the fully expanded form.

The pace of time has meaning only relative to another pace of time. The DeGerlia compactness ratio k_d governs the relative time-dilation between two uniform spherical systems. The standard gravitational time-dilation calculation takes the Schwarzschild form $\sqrt{1-k_s}$, where $k_s = D/D_{crit} = r_s/R$. The DeGerlia compactness ratio $k_d = D_1/D_2$ and the Schwarzschild

form k_s are the same scale factor k evaluated by two routes; the subscript denotes which route produced it.

Via the Schwarzschild radius, the same scale factor is:

$$k_s = \frac{D}{D_{crit}} = \frac{r_s}{R} \quad (5)$$

where $D = M/R$ is the system's DeGerlia Compactness and $r_s = 2GM/c^2$ is its Schwarzschild radius. k_s is the Schwarzschild-route name for k that appears in the standard $\sqrt{1-k}$ for GTD.

3.1 Relating Linear Scale Factor k to General Relativity

Time dilation is always:

$$\frac{d\tau}{dt} = \sqrt{1-k} \quad (6)$$

where $d\tau/dt$ is the time-dilation factor between the two systems and k is the DeGerlia compactness ratio for the case being computed. The specific forms are:

- $k_d = D_1/D_2$ (Equation 4) — via inertial density ratio
- $k_s = D/D_{crit} = r_s/R$ (Equation 5) — via Schwarzschild radius

Same k symbol across forms: a given numerical k produces the same time-scale-factor result regardless of which form it appears in.

3.2 Classical Static Density Example

Setup. A uniform sphere of mass 1 kg and radius 1 m (System 2) is scaled isometrically by a factor of two, producing a sphere of mass $2^3 = 8$ kg and radius 2 m (System 1). The linear scale factor between them is $k = R_1/R_2 = 2$ by construction.

Change in inertia. The moments of inertia are

$$I_1 = \frac{2}{5}(8)(2)^2 = 12.8 \text{ kg m}^2, \quad I_2 = \frac{2}{5}(1)(1)^2 = 0.4 \text{ kg m}^2,$$

so $I_1/I_2 = 32 = 2^5 = k^5$ — the moment of inertia changes by the fifth power of the linear scale factor.

Gravitational time dilation. Each system's Schwarzschild radius is $r_s = 2GM/c^2$, giving $r_{s,1} = 1.18819 \times 10^{-26} \text{ m}$ and $r_{s,2} = 1.48523 \times 10^{-27} \text{ m}$. The Schwarzschild gravitational time dilation $gtd = \sqrt{1-r_s/R}$ is then

$$gtd_1 = \sqrt{1 - \frac{r_{s,1}}{R_1}} = 1 - 2.97046 \times 10^{-27},$$

$$gtd_2 = \sqrt{1 - \frac{r_{s,2}}{R_2}} = 1 - 7.42616 \times 10^{-28},$$

so the relative time dilation between the two systems is

$$\frac{gtd_1}{gtd_2} = 1 - 2.22785 \times 10^{-27}.$$

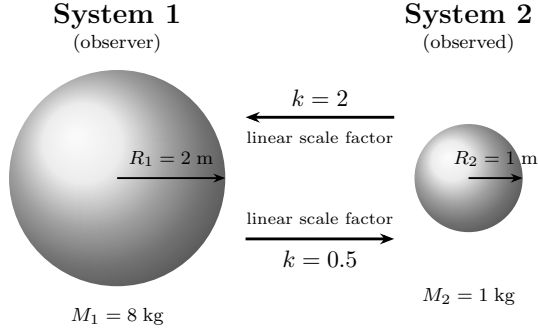
DeGerlia compactness ratio. The ratio k_d follows from the DeGerlia compactness $D = M/R$ ($D_1 = 4 \text{ kg/m}$, $D_2 =$

* This identity has been previously referred to as the "space-time equivalence"⁶.

1 kg/m):

$$k_d = \frac{D_1}{D_2} = \frac{4}{1} = 4 = k^2, \quad \sqrt{k_d} = 2 = k.$$

This matches the Classical Static Density Example column of Appendix Appendix A: the gravitational time dilation ratio is $1 - 2.22785 \times 10^{-27}$ and $\sqrt{k_d} = 2$.



Two uniform spheres related by isometric scaling, scale factor $k = 2$: System 1 ($M_1 = 8 \text{ kg}$, $R_1 = 2 \text{ m}$) and System 2 ($M_2 = 1 \text{ kg}$, $R_2 = 1 \text{ m}$). The moment of inertia changes by $2^5 = 32$; the relative time dilation $1 - 2.22785 \times 10^{-27}$ is obtained equivalently from the Schwarzschild and inertial-density formulations.

4 General and Special Relativity

Schwarzschild gravitational time dilation:

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{r_s}{R}} \quad (7)$$

where $r_s = 2GM/c^2$ is the Schwarzschild radius and R is the radius.

Lorentz time dilation:

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{v^2}{c^2}} \quad (8)$$

where v is the velocity of the moving frame and c is the speed of light.

These are the same form as Equation 6, where $k = r_s/R$ in the gravitational case and $k = v^2/c^2$ in the velocity case. The two expressions are equivalent: the escape velocity at radius R is $v^2 = 2GM/R$, so $v^2/c^2 = 2GM/(Rc^2) = r_s/R$.

Schwarzschild gravitational time dilation is the STE (Equation 4) evaluated under the constraint $M_1 = M_2$ (same mass, different radii). Lorentz time dilation is the STE evaluated under the constraint $R_1 = R_2$ (same radius, different masses). Both are cases of the same DeGerlia-compactness-based relation; the derivations of each are developed in Section 5.

5 Expressions of Relative Inertial Linear Scale

The universal scale factor between any two systems is the DeGerlia compactness ratio k_d (Equation 4), with $k_d = D_1/D_2 = M_1 R_2 / (M_2 R_1)$. Each subsection below applies a specific physical constraint that simplifies k_d , derives the resulting form, and

demonstrates the calculation with a worked numerical example.

Numerical values are reported at six significant figures in e-style notation. Reference constants used throughout: $D_{crit} = 6.73295 \times 10^{26} \text{ kg/m}$, $G = 6.67430 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, $c = 2.99792 \times 10^8 \text{ m/s}$. Example parameter sets match the columns of the pair comparison tables (Appendix Appendix A).

5.1 Static Mass ($M_1 = M_2$)

When the two systems share the same mass, k_d is the ratio of their DeGerlia compactness values:

$$k_d = \frac{M_1 R_2}{M_2 R_1} = \frac{D_1}{D_2}$$

Each system's normalized DeGerlia compactness against the DeGerlia Threshold yields $D/D_{crit} = r_s/R$, recovering the Schwarzschild gravitational time dilation parameter.

This case is the standard gravitational time-dilation calculation; the DeGerlia compactness ratio D_1/D_2 produces the familiar Schwarzschild factor here.

Example (Static Mass): System 1 is the system being measured; System 2 is the same mass at its Schwarzschild radius. S_1 : $M_1 = 2.00000 \times 10^{30} \text{ kg}$, $R_1 = 3.00000 \times 10^3 \text{ m}$. S_2 : $M_2 = M_1$, $R_2 = r_s = 2GM_1/c^2 = 2.97046 \times 10^3 \text{ m}$.

DeGerlia compactness for each system:

$$D_1 = \frac{M_1}{R_1} = \frac{2.00000 \times 10^{30}}{3.00000 \times 10^3} = 6.66667 \times 10^{26} \text{ kg/m}$$

$$D_2 = \frac{M_2}{R_2} = \frac{2.00000 \times 10^{30}}{2.97046 \times 10^3} = 6.73295 \times 10^{26} \text{ kg/m}$$

Normalized to the DeGerlia Threshold:

$$D_{norm,1} = \frac{D_1}{D_{crit}} = \frac{6.66667 \times 10^{26}}{6.73295 \times 10^{26}} = 9.90155 \times 10^{-1}$$

$$D_{norm,2} = \frac{D_2}{D_{crit}} = \frac{6.73295 \times 10^{26}}{6.73295 \times 10^{26}} = 1.00000$$

DeGerlia compactness ratio:

$$k_d = \frac{D_1}{D_2} = \frac{D_{norm,1}}{D_{norm,2}} = 9.90155 \times 10^{-1}$$

Schwarzschild ratio:

$$k_s = \frac{r_s}{R_1} = \frac{2.97046 \times 10^3}{3.00000 \times 10^3} = 9.90155 \times 10^{-1}$$

GTd via k_d :

$$\sqrt{1 - k_d} = \sqrt{1 - \frac{D_{norm,1}}{D_{norm,2}}} = 1 - 9.00780 \times 10^{-1}$$

GTd via k_s :

$$\sqrt{1 - k_s} = \sqrt{1 - 9.90155 \times 10^{-1}} = 1 - 9.00777 \times 10^{-1}$$

This case is tabulated in Table 2, column $M = M$ of Appendix Appendix A.

5.2 Static Radius ($R_1 = R_2$)

When the two systems share the same radius, k_d is the ratio of their DeGerlia compactness values:

$$k_d = \frac{M_1 R_2}{M_2 R_1} = \frac{D_1}{D_2}$$

The Lorentz factor (Equation 8) has the same form as Equation 6, with $k = v^2/c^2$. Since velocity is distance per unit time, v^2/c^2 is the square of a ratio of two lengths, with the shared time unit cancelling. Gravitational time dilation confirms this directly: $v^2/c^2 = r_s/R$ via the escape velocity identity, so the velocity-based k of special relativity and the radius-based k of general relativity are the same length ratio.

Example (Static Radius): System 1 is the system being measured; System 2 is at its Schwarzschild radius, defining the common radius. S_1 : $M_1 = 2.00000 \times 10^{28}$ kg, $R_1 = 2.97046 \times 10^3$ m. S_2 : $M_2 = 2.00000 \times 10^{30}$ kg, $R_2 = R_1 = r_s = 2GM_2/c^2 = 2.97046 \times 10^3$ m.

DeGerlia compactness for each system:

$$D_1 = \frac{M_1}{R_1} = \frac{2.00000 \times 10^{28}}{2.97046 \times 10^3} = 6.73295 \times 10^{24} \text{ kg/m}$$

$$D_2 = \frac{M_2}{R_2} = \frac{2.00000 \times 10^{30}}{2.97046 \times 10^3} = 6.73295 \times 10^{26} \text{ kg/m}$$

Normalized to the DeGerlia Threshold:

$$D_{norm,1} = \frac{D_1}{D_{crit}} = \frac{6.73295 \times 10^{24}}{6.73295 \times 10^{26}} = 1.00000 \times 10^{-2}$$

$$D_{norm,2} = \frac{D_2}{D_{crit}} = \frac{6.73295 \times 10^{26}}{6.73295 \times 10^{26}} = 1.00000$$

DeGerlia compactness ratio:

$$k_d = \frac{D_1}{D_2} = \frac{D_{norm,1}}{D_{norm,2}} = 1.00000 \times 10^{-2}$$

Schwarzschild ratio for System 1 ($r_{s,1} = 2GM_1/c^2 = 2.97046 \times 10^1$ m):

$$k_s = \frac{r_{s,1}}{R_1} = \frac{2.97046 \times 10^1}{2.97046 \times 10^3} = 1.00000 \times 10^{-2}$$

GTD via k_d :

$$\sqrt{1 - k_d} = \sqrt{1 - \frac{D_{norm,1}}{D_{norm,2}}} = 1 - 5.01256 \times 10^{-3}$$

GTD via k_s :

$$\sqrt{1 - k_s} = \sqrt{1 - 1.00000 \times 10^{-2}} = 1 - 5.01256 \times 10^{-3}$$

This case is tabulated in Table 2, column $R = R$ of Appendix Appendix A.

5.3 Constant Density (Isometric) ($\rho_1 = \rho_2$)

When the two systems share the same density, $M \propto R^3$, so $M_1/M_2 = (R_1/R_2)^3$. Substituting into k_d :

$$k_d = \frac{M_1 R_2}{M_2 R_1} = \left(\frac{R_1}{R_2}\right)^3 \cdot \frac{R_2}{R_1} = \left(\frac{R_1}{R_2}\right)^2$$

The DeGerlia compactness ratio reduces to the square of the radius ratio. The linear scale factor follows as $\sqrt{k_d} = R_1/R_2$, recovering the isometric equivalence.

Example (Constant Density): System 1 is the system being measured; System 2 is the comparator at its Schwarzschild radius. S_1 : $M_1 = 7.63040 \times 10^{28}$ kg, $R_1 = 1.00000 \times 10^3$ m. S_2 : $M_2 = 2.00000 \times 10^{30}$ kg, $R_2 = r_s = 2GM_2/c^2 = 2.97046 \times 10^3$ m. Both at $\rho = 1.82181 \times 10^{19}$ kg/m³.

DeGerlia compactness ratio:

$$k_d = \frac{M_1 R_2}{M_2 R_1} = \frac{(7.63040 \times 10^{28})(2.97046 \times 10^3)}{(2.00000 \times 10^{30})(1.00000 \times 10^3)} = 1.13327 \times 10^{-1}$$

Schwarzschild ratio for System 1 ($r_{s,1} = 2GM_1/c^2 = 1.13327 \times 10^2$ m):

$$k_s = \frac{r_{s,1}}{R_1} = \frac{1.13327 \times 10^2}{1.00000 \times 10^3} = 1.13327 \times 10^{-1}$$

GTD via k_d :

$$\sqrt{1 - k_d} = \sqrt{1 - 1.13327 \times 10^{-1}} = 1 - 5.83337 \times 10^{-2}$$

GTD via k_s :

$$\sqrt{1 - k_s} = \sqrt{1 - 1.13327 \times 10^{-1}} = 1 - 5.83337 \times 10^{-2}$$

This case is tabulated in Table 2, column $\rho = \rho$ of Appendix Appendix A.

5.4 Constant Inertia ($I_1 = I_2$)

When the two systems share the same moment of inertia, $M_1 R_1^2 = M_2 R_2^2$, so $M_1/M_2 = (R_2/R_1)^2$. Substituting into k_d :

$$k_d = \frac{M_1 R_2}{M_2 R_1} = \left(\frac{R_2}{R_1}\right)^2 \cdot \frac{R_2}{R_1} = \left(\frac{R_2}{R_1}\right)^3$$

The DeGerlia compactness ratio reduces to the cube of the radius ratio.

Example (Constant Inertia): System 1 is the system being measured; System 2 is the comparator at its Schwarzschild radius. S_1 : $M_1 = 1.76450 \times 10^{29}$ kg, $R_1 = 1.00000 \times 10^4$ m. S_2 : $M_2 = 2.00000 \times 10^{30}$ kg, $R_2 = r_s = 2GM_2/c^2 = 2.97046 \times 10^3$ m. Both have $I = MR^2 = 1.76450 \times 10^{37}$ kg·m².

DeGerlia compactness ratio:

$$k_d = \frac{M_1 R_2}{M_2 R_1} = \frac{(1.76450 \times 10^{29})(2.97046 \times 10^3)}{(2.00000 \times 10^{30})(1.00000 \times 10^4)} = 2.62034 \times 10^{-2}$$

Schwarzschild ratio for System 1 ($r_{s,1} = 2GM_1/c^2 =$

$2.62034 \times 10^2 \text{ m})$:

$$k_s = \frac{r_{s,1}}{R_1} = \frac{2.62034 \times 10^2}{1.00000 \times 10^4} = 2.62034 \times 10^{-2}$$

GTD via k_d :

$$\sqrt{1 - k_d} = \sqrt{1 - 2.62034 \times 10^{-2}} = 1 - 1.31836 \times 10^{-2}$$

GTD via k_s :

$$\sqrt{1 - k_s} = \sqrt{1 - 2.62034 \times 10^{-2}} = 1 - 1.31836 \times 10^{-2}$$

This case is tabulated in Table 2, column $I = I$ of Appendix Appendix A.

5.5 General Case (unconstrained)

When no constraint is applied between the systems, k_d stays in its full form:

$$k_d = \frac{M_1 R_2}{M_2 R_1}$$

Both mass and radius must be specified independently for each system.

Example (General Case): System 1 is the system being measured; System 2 is the comparator at its Schwarzschild radius. S_1 : $M_1 = 1.00000 \times 10^{29} \text{ kg}$, $R_1 = 1.00000 \times 10^4 \text{ m}$. S_2 : $M_2 = 2.00000 \times 10^{30} \text{ kg}$, $R_2 = r_s = 2GM_2/c^2 = 2.97046 \times 10^3 \text{ m}$.

DeGerlia compactness ratio:

$$k_d = \frac{M_1 R_2}{M_2 R_1} = \frac{(1.00000 \times 10^{29})(2.97046 \times 10^3)}{(2.00000 \times 10^{30})(1.00000 \times 10^4)} = 1.48523 \times 10^{-2}$$

Schwarzschild ratio for System 1 ($r_{s,1} = 2GM_1/c^2 = 1.48523 \times 10^2 \text{ m})$:

$$k_s = \frac{r_{s,1}}{R_1} = \frac{1.48523 \times 10^2}{1.00000 \times 10^4} = 1.48523 \times 10^{-2}$$

GTD via k_d :

$$\sqrt{1 - k_d} = \sqrt{1 - 1.48523 \times 10^{-2}} = 1 - 7.45394 \times 10^{-3}$$

GTD via k_s :

$$\sqrt{1 - k_s} = \sqrt{1 - 1.48523 \times 10^{-2}} = 1 - 7.45394 \times 10^{-3}$$

This case is tabulated in Table 2, column General of Appendix Appendix A.

licly available at: https://github.com/denverdata/academic_InertialRelativity-deriving-gr-sr-from-ste.

References

- [1] T. D. DeGerlia, *Introducing Inertial Density and Characterizing the Schwarzschild Condition as a Constant Threshold of Mass over Radius (m/r)*, DEC, 2025, DOI: 10.55277/researchhub.0e0cgoor.
- [2] A. Einstein, *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften (Berlin)*, 1915, 844–847.
- [3] K. Schwarzschild, *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften (Berlin)*, 1916, 189–196.
- [4] A. Einstein, *Annalen der Physik*, 1905, **17**, 891–921.
- [5] E. Tiesinga, P. J. Mohr, D. B. Newell and B. N. Taylor, *Reviews of Modern Physics*, 2021, **93**, 025010.
- [6] T. D. DeGerlia, *The Universe of Light - Exploring our Universe through the Lens of the Law of Space-Time Equivalence*, DEC, 2025, DOI: 10.55277/ResearchHub.pi9yjshm.
- [7] H. Goldstein, C. Poole and J. Safko, *Classical Mechanics*, Addison-Wesley, 3rd edn, 2002, pp. 70–71.

Appendix A Pair Comparison Tables

Tables 1 and 2 present the complete system properties and derived ratios for six system pairs spanning the constraint cases from the derivations in Section 5: static mass ($M=M$), static radius ($R=R$), constant density ($\rho = \rho$), constant inertia ($I=I$), an unconstrained general case, and an STE example. The three algebraic forms of k_d in Equation 4 produce identical values in every case, alongside the gravitational time-dilation ratio computed independently from the standard Schwarzschild form.

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Conflicts of Interest

The author declares no competing interests.

Data Availability

The complete source materials for this paper, including all $\LaTeX$ source files, derivation documents, and revision history, are pub-

Table 1 System properties for each pair comparison case.

	Case 1 ($M=M$)	Case 2 ($R=R$)	Case 3 ($\rho=\rho$)	Case 4 ($I=I$)	General	Classical ($\rho=\rho$)
m_1 (kg)	2.00000e+30	1.00000e+30	1.00000e+30	1.00000e+32	1.00000e+20	8.00000e+0
m_2 (kg)	2.00000e+30	1.00000e+24	1.00000e+27	1.00000e+30	1.00000e+30	1.00000e+0
r_1 (m)	3.00000e+3	1.00000e+8	1.00000e+8	1.00000e+7	2.00000e-7	2.00000e+0
r_2 (m)	7.00000e+8	1.00000e+8	1.00000e+7	1.00000e+8	1.00000e+7	1.00000e+0
ρ_1 (kg/m ³)	1.76839e+19	2.38732e+5	2.38732e+5	2.38732e+10	2.98416e+39	2.38732e-1
ρ_2 (kg/m ³)	1.39203e+3	2.38732e-1	2.38732e+5	2.38732e+5	2.38732e+8	2.38732e-1
I_1 (kg·m ²)	7.20000e+36	4.00000e+45	4.00000e+45	4.00000e+45	1.60000e+6	1.28000e+1
I_2 (kg·m ²)	3.92000e+47	4.00000e+39	4.00000e+40	4.00000e+45	4.00000e+43	4.00000e-1
D_1 (kg/m)	6.66667e+26	1.00000e+22	1.00000e+22	1.00000e+25	5.00000e+26	4.00000e+0
D_2 (kg/m)	2.85714e+21	1.00000e+16	1.00000e+20	1.00000e+22	1.00000e+23	1.00000e+0
$r_{s,1}$ (m)	2.97046e+3	1.48523e+3	1.48523e+3	1.48523e+5	1.48523e-7	1.18819e-26
$r_{s,2}$ (m)	2.97046e+3	1.48523e-3	1.48523e+0	1.48523e+3	1.48523e+3	1.48523e-27
$D_{1,norm}$	9.90155e-1	1.48523e-5	1.48523e-5	1.48523e-2	7.42617e-1	5.94093e-27
$D_{2,norm}$	4.24352e-6	1.48523e-11	1.48523e-7	1.48523e-5	1.48523e-4	1.48523e-27
k_s	9.90155e-1	1.48523e-5	1.48523e-5	1.48523e-2	7.42616e-1	5.94093e-27
k_d	9.90155e-1	1.48523e-5	1.48523e-5	1.48523e-2	7.42617e-1	5.94093e-27
GTD _{s,1}	1-9.00777e-1	1-7.42619e-6	1-7.42619e-6	1-7.45394e-3	1-4.92670e-1	1-2.97046e-27
GTD _{s,2}	1-2.12176e-6	1-7.42616e-12	1-7.42616e-8	1-7.42619e-6	1-7.42644e-5	1-7.42616e-28
GTD _{d,1}	1-9.00780e-1	1-7.42619e-6	1-7.42619e-6	1-7.45395e-3	1-4.92670e-1	1-2.97047e-27
GTD _{d,2}	1-2.12176e-6	1-7.42617e-12	1-7.42617e-8	1-7.42619e-6	1-7.42644e-5	1-7.42617e-28

Table 2 Derived ratios across constraint cases.

	Case 1 ($M=M$)	Case 2 ($R=R$)	Case 3 ($\rho=\rho$)	Case 4 ($I=I$)	General	Classical ($\rho=\rho$)
$\sqrt{D_1/D_2}$	4.83046e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$\sqrt{(m_1 r_2)/(m_2 r_1)}$	4.83046e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$\sqrt{(I_1/I_2)(r_2/r_1)^3}$	4.83046e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$(I_1/I_2)^{1/5}(\rho_1/\rho_2)^{3/10}$	4.83046e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$(m_1 r_2)/(m_2 r_1)$	2.33333e+5	1.00000e+6	1.00000e+2	1.00000e+3	5.00000e+3	4.00000e+0
D_1/D_2	2.33333e+5	1.00000e+6	1.00000e+2	1.00000e+3	5.00000e+3	4.00000e+0
$(I_1/I_2)(r_2/r_1)^3$	2.33333e+5	1.00000e+6	1.00000e+2	1.00000e+3	5.00000e+3	4.00000e+0
GTD _{s,1} /GTD _{s,2}	1-9.00776e-1	1-7.42618e-6	1-7.35193e-6	1-7.44657e-3	1-4.92632e-1	1-2.22785e-27
GTD _{d,1} /GTD _{d,2}	1-9.00780e-1	1-7.42619e-6	1-7.35193e-6	1-7.44658e-3	1-4.92633e-1	1-2.22785e-27
I_1/I_2	1.83673e-11	1.00000e+6	1.00000e+5	1.00000e+0	4.00000e-38	3.20000e+1
$(I_1/I_2)^{1/2}$	4.28571e-6	1.00000e+3	3.16228e+2	1.00000e+0	2.00000e-19	5.65685e+0
$(\rho_1/\rho_2)^{1/5}(r_1/r_2)$	7.12540e-3	1.58489e+1	1.00000e+1	1.00000e+0	3.31445e-8	2.00000e+0
$(I_1/I_2)^{1/5}$	7.12540e-3	1.58489e+1	1.00000e+1	1.00000e+0	3.31445e-8	2.00000e+0
m_1/m_2	1.00000e+0	1.00000e+6	1.00000e+3	1.00000e+2	1.00000e-10	8.00000e+0
$\sqrt{m_1/m_2}$	1.00000e+0	1.00000e+3	3.16228e+1	1.00000e+1	1.00000e-5	2.82843e+0
r_1/r_2	4.28571e-6	1.00000e+0	1.00000e+1	1.00000e-1	2.00000e-14	2.00000e+0
$\sqrt{r_1/r_2}$	2.07020e-3	1.00000e+0	3.16228e+0	3.16228e-1	1.41421e-7	1.41421e+0